

Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Pre-Tutorial Assignment - 10

Date: Nov 2, 2023

Marks: 2

1. For the emitter-bias network in the active region (base-emitter forward bias, base-collector reverse bias) Fig.1. Let $V_{CC} = +20V$, $\beta = 50$, $R_B = 430k\Omega$, $R_C = 2k\Omega$, $R_E = 1k\Omega$. Determine the following (assume $V_{BE} = 0.7V$):

- (a) I_B
- (b) I_C
- (c) V_{CE}

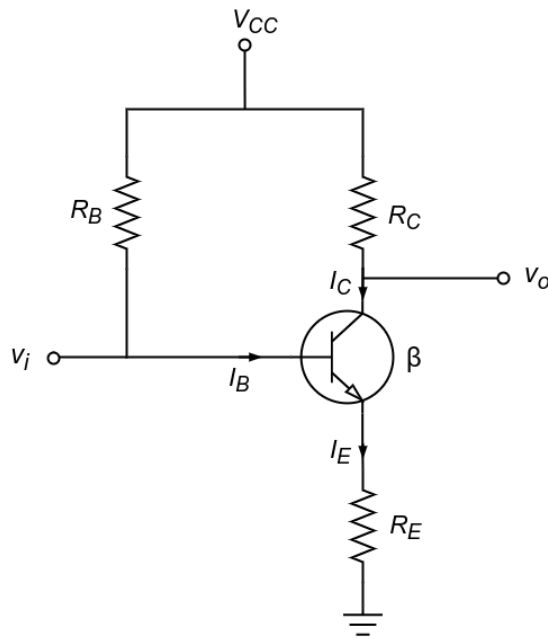


Figure 1: